TINGLONG FENG

Institute for Theoretical Physics ⋄ Utrecht University
https://arendelle-ftl.github.io/ ⋄ t.feng@students.uu.nl

EDUCATION

Utrecht University, Master of Science in Theoretical Physics — September 2024 - July 2026
GPA: 8.14/10
Xi'an Jiaotong University, Bachelor of Medicine in Clinical Medicine — September 2019 - July 2024
GPA for **advanced mathematics and physics courses**: 3.89/4.30 or 91.48/100, including

Main Courses	Credits	Grades	Main Courses	Credits	Grades
Mathematical and Physical Equation	2	98	Electrodynamics	4	87
Complex Analysis and Integral Transformation	3	96	Quantum Field Theory	3	A
Thermodynamics and Statistical Physics I	4	94	Introductory General Relativity	2	A
Quantum Mechanics	4	89	Introduction to Group Theory	2	92
Theoretical Mechanics	3	89	Introduction to Elementary Particles	2	84

PUBLICATIONS

1 **T. Feng**, J. Moes, and T. Prokopec, "Dawn and Twilight Time in Quantum Tunneling," *arXiv*:2512.14809 [quant-ph] (2025).

2 Y. Bai, **T.-L. Feng**, S. Kim, C.-Y. Lee, L.-H. Liu, W. Zhao, and S. Zhou, "Correlators for pseudo Hermitian systems," *JHEP* **11** (2024) 161. DOI: 10.1007/JHEP11(2024)161. (arXiv:2408.07506)

3 **T. Feng**, "Stability of the Potential Super Jupiter in Alpha Centauri System," *arXiv*:2406.19177 [astro-ph.EP] (2024).

4 **T. Feng**, "Holographic Approach to Neutron Stars," *arXiv*:2401.01617 [hep-th] (2024).

ACADEMIC SCHOOLS AND CONFERENCES

DRSTP PhD School and Symposium: Trends in Theory 2026 — May 18 - 22, 2026

· MSc Fellowship for the PhD School; **poster** presentation at the Symposium, Wageningen.

Jürgen Ehlers Spring School on Gravitational Physics 2026 — February 23 - March 06, 2026

· Selected participant, Max Planck Institute for Gravitational Physics (Albert Einstein Institute), Potsdam.

PROGRAMMING

Familiar with C++, MATLAB, Mathematica, Python

RESEARCH PROJECTS

Spherical Freezing Analogy for Cosmology — January 2026 - present

Supervisor: Prof. Federico Toschi, Eindhoven University of Technology

· Developed a spherical freezing model as an exact analogue of radiation-era FLRW expansion.
· Formulated an axisymmetric extension incorporating buoyancy-driven convection, linking interface deformation to shear-like dynamics in Bianchi-type cosmologies.
· Independently learned and used AFiD-MuRPhFi phase-field simulations to test Stefan-type growth laws and analyse interface deformation modes.

A Study of Quantum Tunneling — September 2025 - present

Supervisor: Prof. Tomislav Prokopec, Utrecht University

· Developed a real-time, flux-based analytic framework for metastable decay, capturing short-time deviations, exponential decay, and late-time power-law tails.
· Derived closed-form, computable *dawn* and *twilight* time scales and applied them to 1D models.

Correlators for pseudo Hermitian systems — January-August 2024

Supervisor: Prof. Siyi Zhou, Chongqing University

- Developed the in-in and Schwinger–Keldysh formalisms for pseudo-Hermitian fields.
- Analyzed loop corrections to primordial non-Gaussianity. Indicated the symplectic-fermion loop differs from the scalar-boson case by a relative minus sign

Stability of Planetary System — June 2024

Course project for Theoretical Mechanics, Xi'an Jiaotong University

- Deployed REBOUND/MEGNO framework to investigate long-term orbital stability in binary systems.
- Identified regions of phase space that could host a stable Jupiter-mass planet around Alpha Centauri AB. Compared them with analogous configurations like the Neptune-mass planet in GJ 65AB.

Graphene Production — September 2019 - May 2021

Supervisor: Prof. Wei Wei, Xi'an Jiaotong University

- Innovated traditional electrochemical exfoliation methods to produce high-quality graphene. Achieved better ion intercalation by modifying an electrolytic cell with graphite sheets and employing an electric field.
- Confirmed less defective graphene compared to conventional methods using physical modeling and Raman spectroscopy

Health Economics Modelling of Infectious Disease — May 2020 - May 2021

Supervisor: Prof. Fan Zhang, Xi'an Jiaotong University

- Employed mathematical modeling to analyze government funding allocation between clinical treatment and preventive medicine during epidemics
- Suggested prioritizing clinical treatment for diseases like influenza with high prevalence and recovery rates, while prioritizing preventive medicine for low prevalence and recovery rate diseases like cancer.

TEACHING

Host of Theoretical Physics Seminars

QFT Seminars (website, recording) — November 2025

Cosmology Seminars (recording) — September 2024

Astrophysics Seminars (website, recording) — March 2024

Quantum Mechanics Seminars (website, recording) — April 2021

Classical Electrodynamics Seminars (website, recording) — February 2021

Analytical Mechanics Seminars (website, recording) — April 2020

SERVICE AND OUTREACH

United Academic Forum of Basic Science for Undergraduates — December 2020 - February 2022

- Developed and co-organized this communication platform designed for undergraduates from various universities, fostering collaboration and engagement in basic science research. See this website.

Student Organisation at Xi'an Jiaotong University

- Staff of Scholastic Guidance Center of Qian Xuesen Honors College — September 2020 - August 2021
- Deputy editor-in-chief of *Journal of Zhufeng* — September 2021 - February 2022
- Reviewer of *Journal of Zhufeng* — September 2020 - present

Student Organisation at Utrecht University

- Member of ABBA Student Association — June 2025 - present

LANGUAGE

Fluent in English (CEFR C1), capable of speaking a little Quenya (my Quenya learning notes)